CALIBER 2026 | CASE 1: MANUFACTURING KNOWLEDGE HUB

Business Impact and Financial Modelling Report

Asset Knowledge Platform

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IDR 3.8 B
annual value at full rollout, 3 units

IDR 3.3 B
net present value before tax, 3 units

34 months
payback, 3 units

82%
probability of a positive NPV

Version 1.0 | 3 October 2026 | Companion to Business_Impact_Financial_Model.xlsx. Base scenario. Sample data provided for CALIBER purposes only.

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1 Executive summary

> The platform pays back when it is rolled out beyond one unit. A three-unit programme returns about IDR 3.8 B a year and a positive net present value in 82% of simulated outcomes.

This report rebuilds the business impact slide of the Asset Knowledge Platform proposal as a full financial model. Every value on the slide now traces to a formula in the accompanying Excel workbook, to the 211 work orders supplied with the case, or to a cited market source. The workbook has 23 sheets, about 2,187 live formulas and 28 integrity checks, and a separate Python replica reproduces its results to the last digit.

The Base scenario counts only four evidence-based levers: time spent searching and checking sources, repair time, repeat failures and faster start for new staff. Start-up delay, incident cost and document rework are reported as upside and are not part of any headline figure.

Base scenario	1 unit (pilot)	3 units	6 units	12 units
Annual value at full rollout (IDR M, year 0 prices)	1,275	3,825	7,458	14,533
One-off cost over the programme (IDR M)	1,766	2,708	4,148	7,067
Net present value before tax (IDR M)	-205	3,320	7,446	14,131
Net present value after tax (IDR M)	-252	2,442	5,560	10,550
Internal rate of return before tax	8%	44%	62%	77%
Payback (months)	50	34	32	34
Probability that NPV is positive (Monte Carlo)	27%	82%	93%	97%

1.1 Main findings

- **The pilot unit does not repay on its own.** One unit returns IDR 1.3 B a year against a one-off cost of IDR 1.8 B, so its NPV is IDR -0.2 B. The pilot buys evidence and a reusable platform.
- **Scale is what creates value.** The shared cost of IDR 1.3 B is paid once. NPV turns positive from two units and reaches IDR 14.1 B at twelve units. Chandra Asri has about 21 unit equivalents, so twelve units is 58% of its scale.

- **The risk is bounded.** In 5,000 simulated outcomes the three-unit NPV has a 5th percentile of IDR -1.4 B and a 95th percentile of IDR 7.6 B. The largest drivers are the share of breakdown hours that reduce output, the margin per tonne and the pay level used to value time.
- **The market is a value pool, not a revenue pool.** Indonesian chemical and petrochemical manufacturing holds about 480 process units and a customer value pool of IDR 611.9 B a year. A vendor could capture only about IDR 122,387 M (6.8 million USD), so the case rests on internal value.
- **Several earlier values were too optimistic and have been corrected.** The cost anchor, the value of a downtime hour, the share of freed time that is productive and the share of the gain credited to the platform are now explicit.

Earlier slide	Earlier value	New model	Reason
Annual value, 1 unit (IDR M)	1,521	1,275	Haircuts for productive time and attribution; adoption of 85 percent
Pilot one-off cost (IDR M)	1,400	1,766	Rebuilt line by line
Pilot running cost (IDR M a year)	370	731	Knowledge manager, platform engineer, hosting, security
Payback, 3 units (months)	12	34	Phased rollout and adoption ramp by cohort
Value of a breakdown hour (IDR M)	30	33.1	Built from capacity, margin and share of hours with output loss

2 Context, scope and method

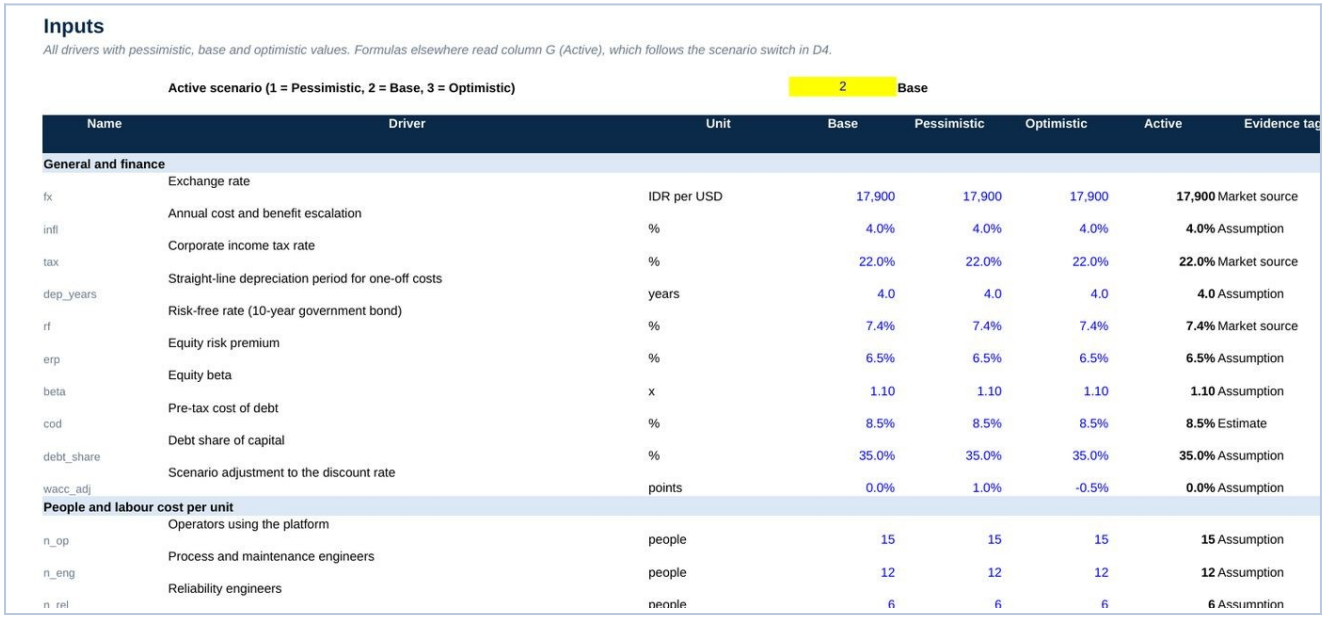
The CALIBER 2026 Case Book asks for a Manufacturing Knowledge Hub that connects scattered plant knowledge, gives engineers trusted and cited answers and supports reliability decisions. The business impact section must show expected benefits, key performance indicators and measurable value with supporting evidence. This report answers the business side of the three key questions.

Key question	Business answer in this report
1. How can the Company build a structured Industrial Data Ops foundation?	A one-off shared cost of IDR 1,322 M covers four connectors, security and governance. Section 5 prices it and Section 6 shows that it is repaid from two units.
2. How can AI help engineers find trusted information faster?	Search and checking time falls by 50 percent, from 40 to 20 minutes a day. Lever 1 values it at IDR 340 M a year for one unit.
3. How can the hub support reliability and continuous improvement?	Failure memory shortens repairs and prevents repeats. Levers 2 and 3 are 70% of the value.

2.1 Method

1. **Baseline.** Re-derive breakdowns, downtime, repeat failures and cost from the 211 work orders, using formulas on the raw data.
2. **Value.** Convert time saved, repair hours saved and failures avoided into IDR with explicit loaded labour cost and a bottom-up value of a downtime hour.
3. **Cost.** Build one-off, per-unit and running cost line by line, with contingency.
4. **Return.** Phase the rollout by cohort, apply an adoption ramp and discount at the weighted average cost of capital of 11.8%. Report NPV, IRR, payback and benefit-cost ratio, before and after tax.
5. **Stress.** Run one-at-a-time sensitivities, break-even values, a Monte Carlo simulation and a risk register.
6. **Verify.** Recalculate the workbook, check it with 28 integrity tests and compare it with an independent Python replica.

Every input carries an evidence tag: **Case data**, **Market source**, **Estimate** or **Assumption**. The Inputs sheet lists the pessimistic, base and optimistic value of each driver, and one switch changes the whole workbook.



Inputs

All drivers with pessimistic, base and optimistic values. Formulas elsewhere read column G (Active), which follows the scenario switch in D4.

Active scenario (1 = Pessimistic, 2 = Base, 3 = Optimistic) 2 **Base**

Name	Driver	Unit	Base	Pessimistic	Optimistic	Active	Evidence tag
General and finance							
fx	Exchange rate	IDR per USD	17,900	17,900	17,900	**17,900**	Market source
infl	Annual cost and benefit escalation	%	4.0%	4.0%	4.0%	**4.0%**	Assumption
tax	Corporate income tax rate	%	22.0%	22.0%	22.0%	**22.0%**	Market source
dep_years	Straight-line depreciation period for one-off costs	years	4.0	4.0	4.0	**4.0**	Assumption
rf	Risk-free rate (10-year government bond)	%	7.4%	7.4%	7.4%	**7.4%**	Market source
erp	Equity risk premium	%	6.5%	6.5%	6.5%	**6.5%**	Assumption
beta	Equity beta	x	1.10	1.10	1.10	**1.10**	Assumption
cod	Pre-tax cost of debt	%	8.5%	8.5%	8.5%	**8.5%**	Estimate
debt_share	Debt share of capital	%	35.0%	35.0%	35.0%	**35.0%**	Assumption
wacc_adj	Scenario adjustment to the discount rate	points	0.0%	1.0%	-0.5%	**0.0%**	Assumption
People and labour cost per unit							
n_op	Operators using the platform	people	15	15	15	**15**	Assumption
n_eng	Process and maintenance engineers	people	12	12	12	**12**	Assumption
n_rel	Reliability engineers	people	6	6	6	**6**	Assumption

Figure 1. Inputs sheet (extract): blue cells are inputs, the Active column follows the scenario switch, and each driver carries an evidence tag.

3 Market analysis

> The Company's own economics, Indonesian demand and the technology market all point in the same direction: availability and trusted knowledge are worth more each year.

3.1 The Company and the Indonesian market

Indicator	Value	Source
Polyethylene capacity at Cilegon	736 kt a year (336 kt existing and 400 kt added in 2020)	Chemical Engineering; Chandra Asri
Ethylene capacity	860 kt a year, the only producer in Indonesia	Chandra Asri investor update, 2018
Employees, 2024	3,746	Sustainability report 2024
Revenue and result, FY2024	USD 1.79 billion, down 17.4 percent; net loss of USD 69 million	IDNFinancials
Polyethylene and polypropylene demand	3.6 Mt a year, rising about 50 percent to 5.3 Mt by 2027; half is imported	AHK Indonesia
Domestic polyethylene capacity	1.2 Mt a year at about 80 percent utilisation; import quotas since March 2024	AHK Indonesia; Argus
Chemical industry, 2024	Exports of USD 17.4 billion and investment of IDR 65.8 trillion	Ministry of Industry
Southeast Asia LLDPE price, 2026	USD 920 to 1,500 per tonne CFR	Polymer Update

Three conclusions follow. First, Chandra Asri reported a loss in FY2024 in a volatile margin environment, so a digital programme must repay quickly and be released in stages. Second, Indonesia imports about half of its polyethylene

and is raising import barriers, so every point of domestic utilisation matters: one percentage point on 1.2 Mt of capacity is 12 kt, or about USD 2.6 million of margin at USD 220 per tonne. Third, plant knowledge is at risk: surveys report that about 40 percent of chemical workforces are aged 45 or over and that most knowledge is tacit.

3.2 Technology and services market

- Indonesia's AI market is estimated at EUR 2.2 billion in 2024 and EUR 10.2 billion in 2030, a growth rate of 28.9 percent a year (Statista).
- Global knowledge management software was USD 20.2 billion in 2024 and grows 13.6 percent a year (Grand View Research). Asset performance management was about USD 24.8 billion.
- Asia Pacific predictive maintenance is USD 3.8 billion in 2026 and reaches USD 7.5 billion in 2031, a growth rate of 14.4 percent a year.
- Knowledge workers spend 1.8 hours a day searching for information (McKinsey) and up to 2.5 hours (IDC). The model uses only 40 minutes a day for plant staff.

3.3 Sizing the opportunity

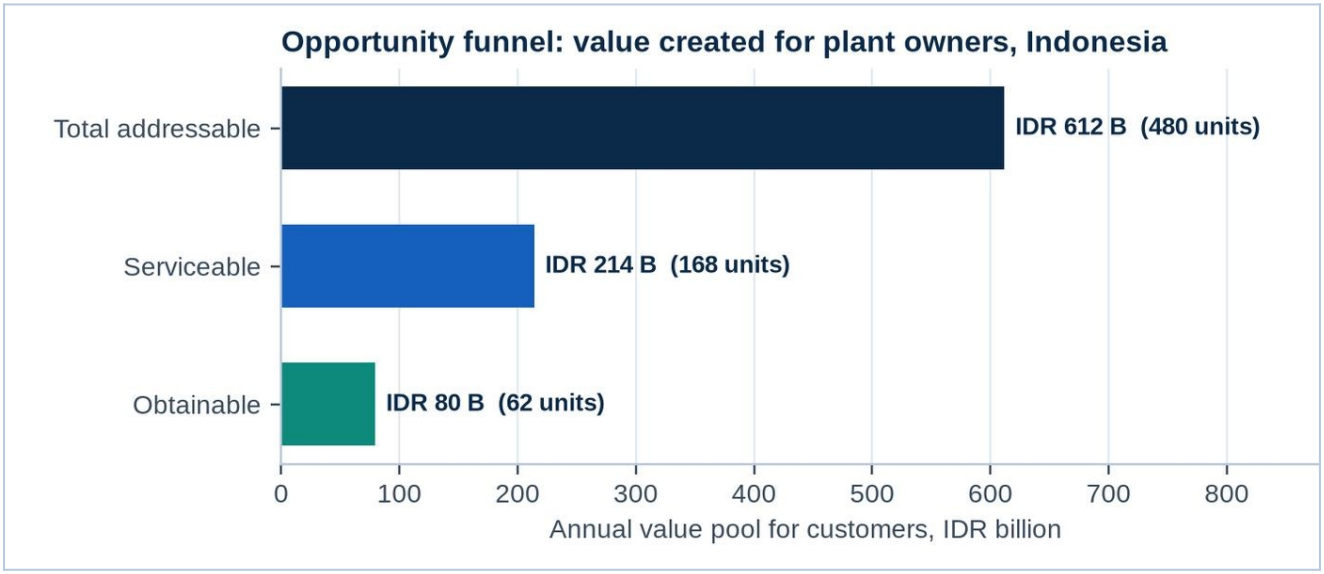


Figure 2. Opportunity funnel for Indonesian chemical and petrochemical manufacturing. Source: Market sheet of the workbook.

The bottom-up sizing assumes 80 manufacturing sites with 6 process units each, which gives 480 units. A unit is a process area the size of the sample LLDPE unit, with about 45 plant staff. Of these, 35 percent sit at sites with an existing document or maintenance system and a digital programme (168 units), and 62 could be reached in five years: 12 internal units and about 50 external ones. At the value per unit of IDR 1,275 M a year, the value pool falls from IDR 611.9 B to IDR 214.2 B and IDR 79.6 B. If a provider priced at 20 percent of value created, the Indonesian revenue pool would be IDR 122,387 M (6.8 million USD), or about 27 million USD with a Southeast Asia multiple of four. A top-down estimate from the national AI market gives a serviceable pool within a factor of 8 of this figure.

> **NOTE**
> The external revenue pool is small. The platform should be funded for the value it creates inside the Company. Commercial extension is an option, not the investment case.

3.4 Alternatives and positioning

Alternatives and positioning

Team assessment on a 1 to 5 scale (5 is best). Scores are judgements to be tested with the Company. Only the list price of the generic assistant is a published figure.

Criterion	Weight	This platform	Generic workplace assistant	Enterprise document search	EAM or CMMS vendor module	In-house custom bui
Cited answers with approval status and confidence	20%	5	3	2	3	3
Tacit knowledge capture and approval workflow	15%	5	2	1	2	3
Failure memory linked to work orders and lessons	15%	5	1	1	4	3
Plant context: P&ID, interlock, drawing, 3D	15%	5	1	2	3	3
Control of data and deployment	10%	4	2	4	4	5
Time to value	10%	4	4	3	2	1
Five-year cost (5 is lowest)	10%	4	3	4	2	2
Independence from one vendor	5%	4	2	3	2	4
Weighted score	**100%**	**4.65**	**2.20**	**2.25**	**2.85**	**2.95**
Rank		1	5	4	3	2

Figure 3. Weighted comparison of alternatives, Competitors sheet. Scores are the team's assessment on a scale of 1 to 5.

A generic workplace assistant is the closest low-cost alternative. At the published list price of USD 30 per user a month, 135 users cost about IDR 4,350 M over five years before the base Microsoft 365 licence. That is lower than the IDR 8,900 M five-year cost of the platform, but a generic assistant does not hold approval status, interlock logic, work order history or tacit knowledge as structured, auditable facts. Those features create most of the value in Levers 2 and 3.

4 Business impact model

> The model starts from what the work orders show and values four effects. Each step is a formula that can be checked.

4.1 Baseline from the case data

Case metrics

Every figure is a formula on Case_Data. These are the only values taken directly from the case data in the business impact model.

Metric	Value	Definition
History start (first report date)	2024-06-04	Earliest work order report date.
History end (platform cut-off)	2025-12-10	Cut-off used by the platform.
Days covered	554	End minus start.
Years covered	1.52	Days divided by 365.
Work orders	211	All work orders in the file.
Breakdowns	31	Breakdown flag equal to Yes.
Breakdown downtime (hours)	434	Sum of downtime hours on breakdown work orders.
Mean time to repair (hours)	14.0	Downtime hours per breakdown.
Breakdowns a year	20.4	Annualised.
Breakdown downtime a year (hours)	286	Breakdowns a year times MTTR.
Corrective or breakdown jobs	63	Work type Corrective, or breakdown flag Yes.
Corrective or breakdown jobs a year	41.5	Annualised.
Breakdown share of those jobs	49.2%	Breakdowns divided by corrective or breakdown jobs.
Jobs that repeat within 120 days	13	Same equipment tag and same failure mode as an earlier job in the previous 120 days.
Repeat rate	20.6%	Repeating jobs divided by corrective or breakdown jobs.
Average direct cost of a corrective or breakdown job (IDR M)	6.99	Labour plus material cost.

Figure 4. Case_Metrics sheet: every figure is a formula on the 211 work orders.

Between June 2024 and December 2025 the plant recorded 31 breakdowns that cost 434 hours of downtime. The mean time to repair is 14.0 hours and breakdowns run at 20.4 a year, which is 286 downtime hours a year. Of 63 corrective or breakdown jobs, 13 (21%) repeat the same failure mode on the same equipment within 120 days. A direct repair job costs IDR 7.0 million on average. In addition, 66 of the 98 controlled documents carry no approval status.

4.2 Value of one hour of downtime

Value of one hour of unit downtime

Replaces the earlier flat assumption of IDR 30 M an hour with a bottom-up calculation from plant capacity and margin.

Step	Value	Unit	Basis
Unit capacity	336	kt a year	Chandra Asri existing pol
Operating hours a year	8,000	hours	Assumption.
Output per operating hour	42.0	t an hour	Capacity divided by oper
Contribution margin lost per tonne	220	USD per t	Polyethylene price less v
Margin lost per hour of full stoppage	9,240	USD an hour	Output per hour times ma
Exchange rate	17,900	IDR per USD	Databoks, 25 Sep 2026.
Margin lost per hour of full stoppage	165.4	IDR M an hour	Converted to IDR million.
Share of breakdown hours that reduce output	20%	%	Spares, standby units an
Value of one breakdown hour	33.1	IDR M an hour	Used for repair time, repa

Figure 5. Downtime_Value sheet.

The earlier slide used a flat IDR 30 million an hour. The new value starts from a unit capacity of 336 kt a year, which gives 42 tonnes an hour, and a lost margin of USD 220 per tonne. A full stoppage then costs IDR 165 million an hour. Because many equipment sets have spares or only reduce performance, only 20% of breakdown hours are assumed to reduce output, which gives IDR 33.1 million. This is the single most influential assumption in the model and it is tested in Section 7.

4.3 Four value levers

Value levers per unit

Annual value of one unit at 100 percent adoption, in IDR M at year 0 prices. Adoption, rollout and escalation are applied on the cash flow sheets.

Calculation step	Value	Unit	Evidence	Basis
LEVER 1: SEARCH AND CHECKING TIME				
Users per unit	45	people	Assumption	Labour sheet.
Minutes saved per user a day	20.0	minutes	Assumption	Baseline minutes times reduction.
Hours saved per user a year	76.7	hours	Derived	Minutes saved times working days.
Average loaded cost (IDR an hour)	193,500	IDR	Estimate	Labour sheet.
Share of freed time that becomes productive	60%	%	Assumption	Conversion factor.
Lever 1 value (IDR M a year)	**400.5**	IDR M	Assumption	Users times hours times cost times conversion.
LEVER 2: REPAIR TIME				
Breakdowns a year	20.4	events	Case data	Case_Metrics.
Repeat breakdowns avoided a year	1.26	events	Derived	Breakdowns times repeat rate times reduction.
Remaining breakdowns a year	19.16	events	Derived	Avoided repeats are removed first so that lever 3 is not counted twice.
Mean time to repair (hours)	14.0	hours	Case data	Case_Metrics.
Reduction in repair time	10.0%	%	Assumption	Input.
Share credited to the platform	70%	%	Assumption	Input.
Value of one breakdown hour (IDR M)	33.1	IDR M	Estimate	Downtime_Value.
Lever 2 value (IDR M a year)	**621.1**	IDR M	Assumption	Remaining breakdowns times MTTR times reduction times credit times hour value.
LEVER 3: REPEAT FAILURES				
Repeat rate of corrective jobs	20.6%	%	Case data	Case_Metrics.
Corrective jobs a year	41.5	jobs	Case data	Case_Metrics.
Repeat jobs avoided a year	2.57	jobs	Derived	Jobs times repeat rate times reduction.
Of which breakdowns	1.26	events	Derived	Same breakdowns as in lever 2.
Downtime avoided (IDR M)	585.5	IDR M	Derived	Avoided breakdowns times MTTR times hour value.
Direct repair cost avoided (IDR M)	18.0	IDR M	Case data	Avoided jobs times average cost.
Lever 3 value (IDR M a year)	**422.4**	IDR M	Assumption	Sum credited to the platform.
LEVER 4: FASTER START FOR NEW AND ROTATED STAFF				
New joiners and role changes a year	6.0	people	Assumption	Input.
Days saved per joiner	10.0	days	Assumption	Input.
Loaded cost of a working day (IDR)	1,548,000	IDR	Estimate	Hourly cost times hours a day.
Lever 4 value (IDR M a year)	**55.7**	IDR M	Assumption	Joiners times days times cost times conversion.
HEADLINE TOTAL AT 100 PERCENT ADOPTION (IDR M a year)	**1,499.8**	IDR M	Mixed	Levers 1 to 4.
Total at steady-state adoption (IDR M a year)	**1,274.9**	IDR M	Assumption	Multiplied by steady-state adoption.
UPSIDE LEVERS (NOT IN THE HEADLINE)				
Upside A: start-up delay avoided (IDR M a year)	231.6	IDR M	Assumption	Events times hours times full-stop hour value times credit.
Upside B: incident cost avoided (IDR M a year)	150.0	IDR M	Assumption	Frequency times cost times reduction.
Upside C: document rework avoided (IDR M a year)	28.4	IDR M	Assumption	Documents times hours times reduction times cost.
Total upside at 100 percent adoption (IDR M a year)	**410.0**	IDR M	Assumption	Not added to the headline.

Figure 6. Levers sheet: annual value of one unit at full adoption, in IDR M at year 0 prices.

Lever	How it is valued	Evidence	IDR M a year
1 Search and checking time	45 users x 20 minutes saved x 230 days x loaded cost x 60% productive	Assumption, cited context	340
2 Repair time	Remaining breakdowns x 14.0 h x 10% x 70% credited x IDR 33 M an hour	Case data and estimate	528

Lever	How it is valued	Evidence	IDR M a year
3 Repeat failures	Avoided repeat jobs (2.6 a year) x repair cost and downtime, 70% credited	Case data	359
4 Faster start for new staff	6 joiners x 10 days x loaded day cost x 60% productive	Assumption	47
Total at 85% adoption			1,275

Avoided repeat breakdowns are removed before the repair time lever is calculated, so no hour is counted twice. The upside levers are start-up delay avoided (IDR 197 M), incident cost avoided (IDR 128 M) and document rework (IDR 24 M) a year per unit. They rest on assumptions only and are excluded from every headline figure.

5 Investment and running cost

Cost model

One-off cost shared by all units, one-off cost for each unit, and running cost. IDR M at year 0 prices.

Item	Value (IDR M)	Unit	Basis
SHARED ONE-OFF COST			
Platform configuration and hardening	440.0	IDR M	Person-months times blended rate.
Source system connectors	400.0	IDR M	Connectors times cost per connector.
Security review and penetration test	150.0	IDR M	Input.
Governance setup and change design	100.0	IDR M	Input.
Training material and demo content	60.0	IDR M	Input.
Subtotal before contingency	**1,150.0**	IDR M	
Contingency	172.5	IDR M	Percentage of subtotal.
Shared one-off cost	**1,322.5**	IDR M	Incurred in year 0.
ONE-OFF COST FOR EACH UNIT			
Document ingestion, tagging and cleansing	147.0	IDR M	Person-days times cost.
Knowledge capture sessions	144.0	IDR M	Sets times sessions times cost.
User training time	34.8	IDR M	Users times hours times average loaded cost.
Validation and acceptance testing	60.0	IDR M	Input.
Subtotal before contingency	**385.8**	IDR M	
Contingency	57.9	IDR M	Percentage of subtotal.
One-off cost for each unit	**443.7**	IDR M	Incurred the year before the unit goes live.
AI SERVICE USAGE COST			
Questions a year per unit	82,800	queries	Users times questions a day times days.
Input tokens a year (millions)	248.4	M tokens	Questions times tokens per question.
Output tokens a year (millions)	49.7	M tokens	Questions times tokens per answer.
Service cost a year (USD)	596	USD	Tokens times list price times buffer.
Service cost a year (IDR M)	**10.7**	IDR M	Converted at the exchange rate.
RUNNING COST A YEAR			
Cloud hosting and storage	80.0	IDR M	Input.
AI platform agreement and data residency	90.0	IDR M	Input.
Knowledge manager	240.0	IDR M	FTE times monthly loaded cost times 12.
Platform engineer	144.0	IDR M	FTE times monthly loaded cost times 12.
Annual security audit	60.0	IDR M	Input.
Monitoring, licences and tools	40.0	IDR M	Input.
AI service usage	10.7	IDR M	From the usage calculation above.
Subtotal before contingency	**664.7**	IDR M	
Contingency	66.5	IDR M	Percentage of subtotal.
Running cost for the first unit	**731.1**	IDR M	Year 0 prices.

Figure 7. Costs sheet: shared one-off, per-unit one-off and running cost, in IDR M at year 0 prices.

The shared one-off cost is IDR 1,322 M including 15 percent contingency. It covers platform configuration (8 person-months), four connectors (EDMS, AIMS, CMMS and the historian), a security review, governance and training material. Each unit then needs IDR 444 M for document ingestion, knowledge capture sessions, user training and acceptance testing. Running cost is IDR 731 M a year for the first unit and IDR 256 M for each extra unit.

The AI service itself costs only about IDR 11 M a year for one unit at list prices, so it is not a cost driver. People, data work and integration are. The pilot one-off cost is IDR 1,766 M, which is 26% above the earlier estimate of IDR 1,400 M, and the running cost is 98% above IDR 370 M. The earlier figures had no line-by-line support.

6 Financial results

Summary of results

Live links to the cash flow sheets. Change the scenario on Inputs!D4 to see the other cases.

Metric (IDR M unless stated)	1 unit (pilot)	3 units	6 units	12 units
Steady-state annual value (year 0 prices)	**1,275**	**3,825**	**7,458**	**14,533**
Total one-off cost	1,766	2,708	4,148	7,067
Running cost in year 5 (year 0 prices)	731	1,243	2,011	3,546
Total benefits over 5 years	6,444	15,212	26,193	45,280
Total costs over 5 years	5,885	8,900	13,026	20,750
Net benefit over 5 years, before tax	559	6,312	13,167	24,530
NPV before tax	**(205)**	**3,320**	**7,446**	**14,131**
NPV after tax	(252)	2,442	5,560	10,550
IRR before tax	8%	44%	62%	77%
IRR after tax	7%	38%	54%	66%
Payback (months)	**49.9**	**33.9**	**32.2**	**33.7**

Figure 8. Summary sheet: results by programme size, Base scenario, IDR M unless stated.

A three-unit programme costs IDR 2,708 M of one-off spend in total, because units two and three are onboarded in years 1 and 2. It reaches IDR 3,825 M of annual value at full rollout and a net present value of IDR 3,320 M before tax and IDR 2,442 M after tax. The internal rate of return is 44% before tax. Simple payback is 34 months and discounted payback 38 months, because adoption builds over two years and the later units start later. The benefit-cost ratio is 1.48.

Cash flow and returns: 3 units

IDR M, nominal, year-end convention. Year 0 is the investment year. Benefits count the four headline levers only. **Scenario: Base**

Line	Cohort	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Units live at year end		-	1	2	3	3	3
Units going live in the year		-	1	1	1	-	-
Adoption-weighted units by cohort (year gone live)							
Cohort of year 1	1	-	0.51	0.72	0.85	0.85	0.85
Cohort of year 2	2	-	-	0.51	0.72	0.85	0.85
Cohort of year 3	3	-	-	-	0.51	0.72	0.85
Cohort of year 4	4	-	-	-	-	-	-
Cohort of year 5	5	-	-	-	-	-	-
Adoption-weighted units		**-**	**0.51**	**1.23**	**2.08**	**2.42**	**2.55**
Escalation factor		1.000	1.040	1.082	1.125	1.170	1.217
Benefits (headline levers)							
Lever 1: search and checking time		-	212	534	938	1,135	1,243
Lever 2: repair time		-	329	828	1,455	1,760	1,927
Lever 3: repeat failures		-	224	563	990	1,197	1,311
Lever 4: faster start for new staff		-	30	74	131	158	173
Total benefits		**-**	**796**	**1,999**	**3,513**	**4,251**	**4,653**
Memo: upside A, start-up delay		-	*123*	*309*	*542*	*656*	*718*
Memo: upside B, incident cost		-	*80*	*200*	*351*	*425*	*465*
Memo: upside C, document rework		-	*15*	*38*	*67*	*81*	*88*
Costs							
Shared one-off cost		1,323	-	-	-	-	-
One-off cost for units going live next year		444	461	480	-	-	-
Total one-off cost		**1,766**	**461**	**480**	**-**	**-**	**-**
Running cost		-	760	1,068	1,398	1,454	1,512
Total cost		**1,766**	**1,222**	**1,548**	**1,398**	**1,454**	**1,512**
Pre-tax cash flow							
Net cash flow before tax		**(1,766)**	**(426)**	**452**	**2,115**	**2,796**	**3,141**
Cumulative net cash flow		(1,766)	(2,193)	(1,741)	375	3,171	6,312
Discount factor		1.000	0.895	0.800	0.716	0.641	0.573
Present value of net cash flow		(1,766)	(381)	362	1,515	1,791	1,800
Cumulative present value		(1,766)	(2,148)	(1,786)	(271)	1,520	3,320
After-tax cash flow							
Depreciation of one-off cost		-	442	557	677	677	235
Taxable income (benefits less running cost less depreciation)		-	(406)	375	1,438	2,119	2,906
Tax (a negative value is a tax shield)		-	(89)	82	316	466	639

Figure 9. CF_3U sheet: cohort adoption, benefits, costs, cash flow, discounting and tax for the three-unit programme.

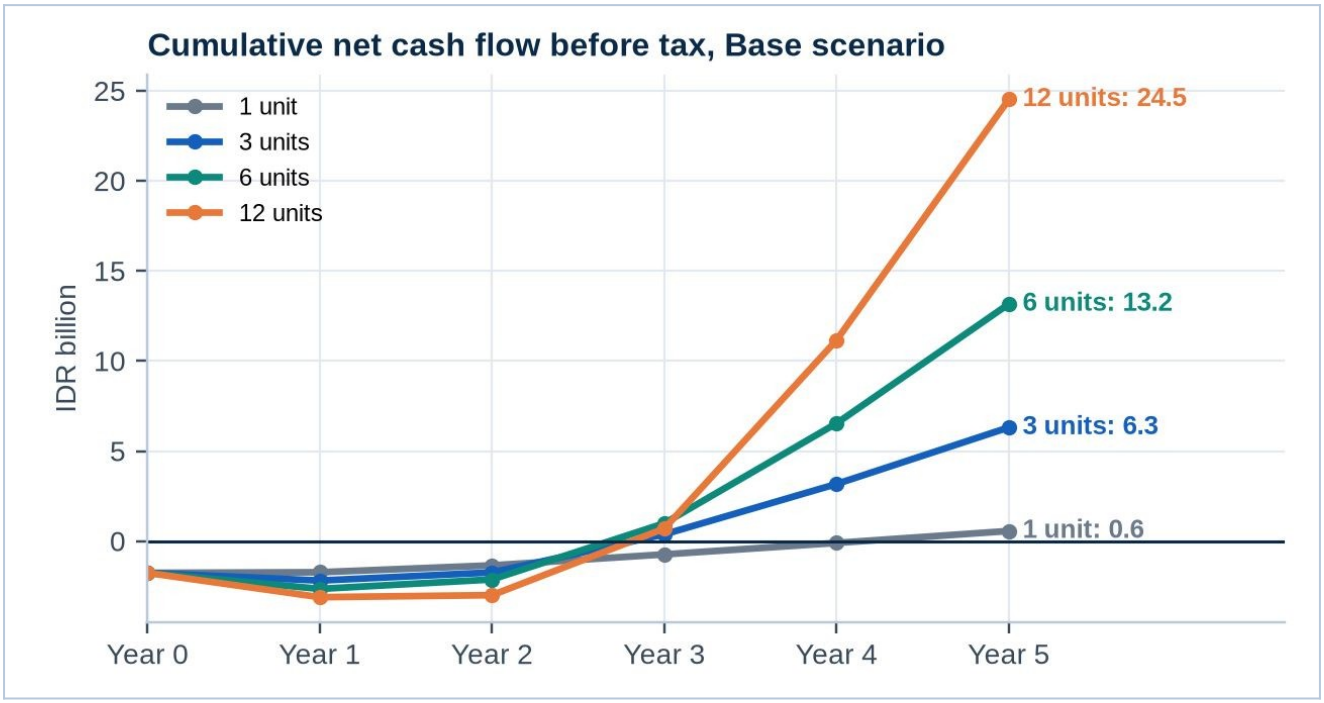


Figure 10. Cumulative net cash flow before tax by programme size, Base scenario.

Cash flow is negative through year 2 in every programme and recovers during year 3. A single unit reaches break-even only after 50 months. Across the scenario switch, the three-unit NPV ranges from IDR -1,299 M in the pessimistic case to IDR 9,974 M in the optimistic case, where every driver moves together. Section 7 shows how likely each outcome is.

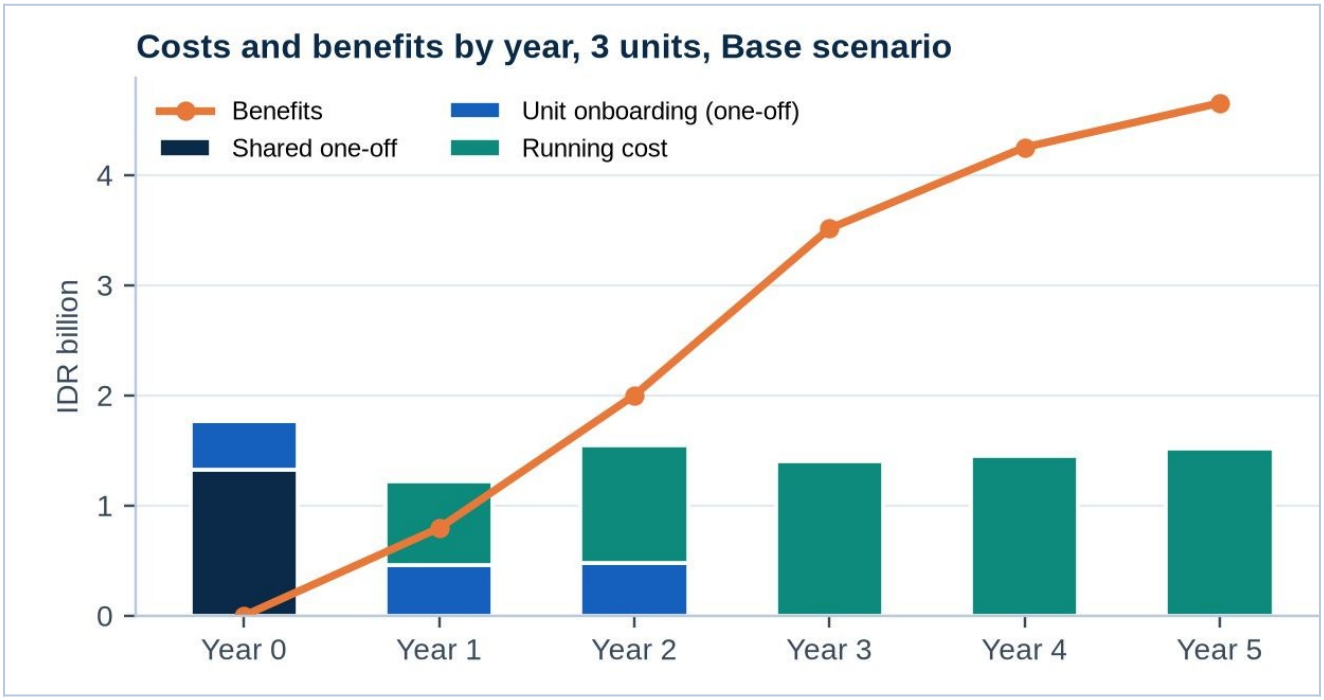


Figure 11. Costs and benefits by year, three units, Base scenario.

7 Uncertainty and risk

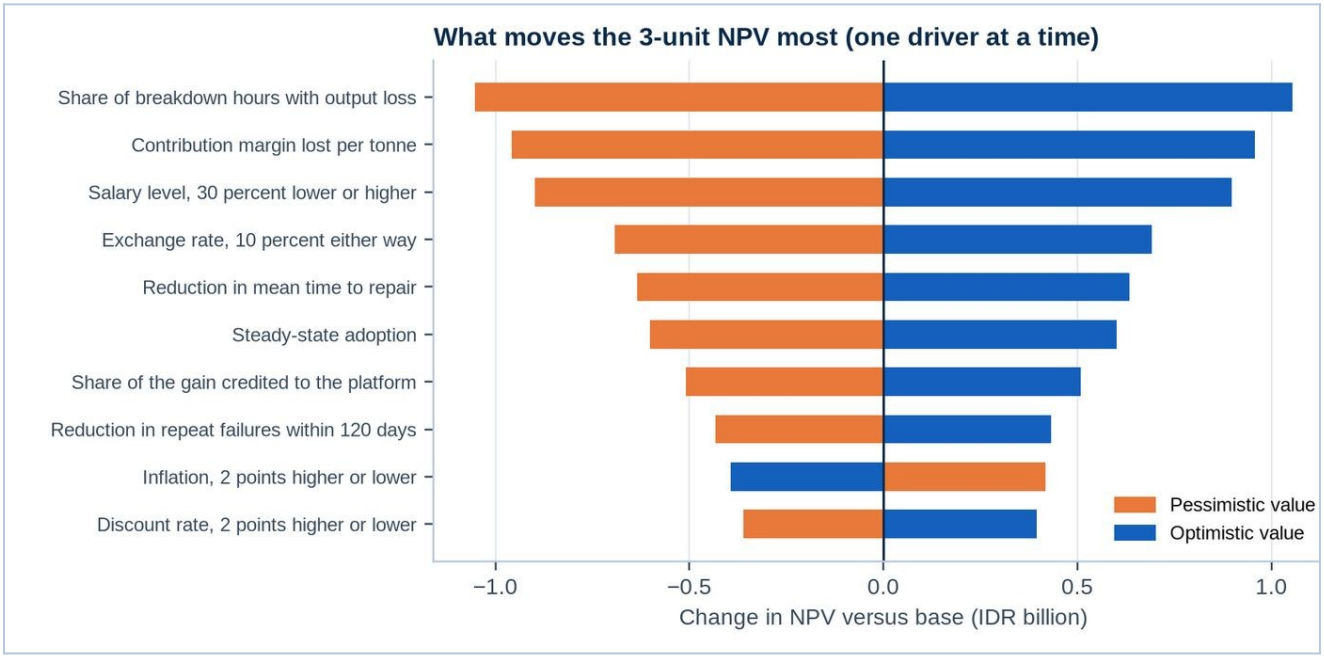


Figure 12. One-at-a-time sensitivity of the three-unit NPV. Source: Sensitivity sheet.

The five largest drivers are: share of breakdown hours that reduce plant output, contribution margin lost per tonne, salary level (all roles), 30 percent lower or higher, exchange rate, 10 percent stronger or weaker rupiah and reduction in mean time to repair. The first two set the value of a downtime hour. The three-unit NPV stays positive until the share of breakdown hours that reduce output falls from 20% to 11%, until the margin falls from USD 220 to USD 116 per tonne, and until the reduction in repair time falls from 10 percent to 2.2%. Costs can rise by 48% before the NPV is zero.

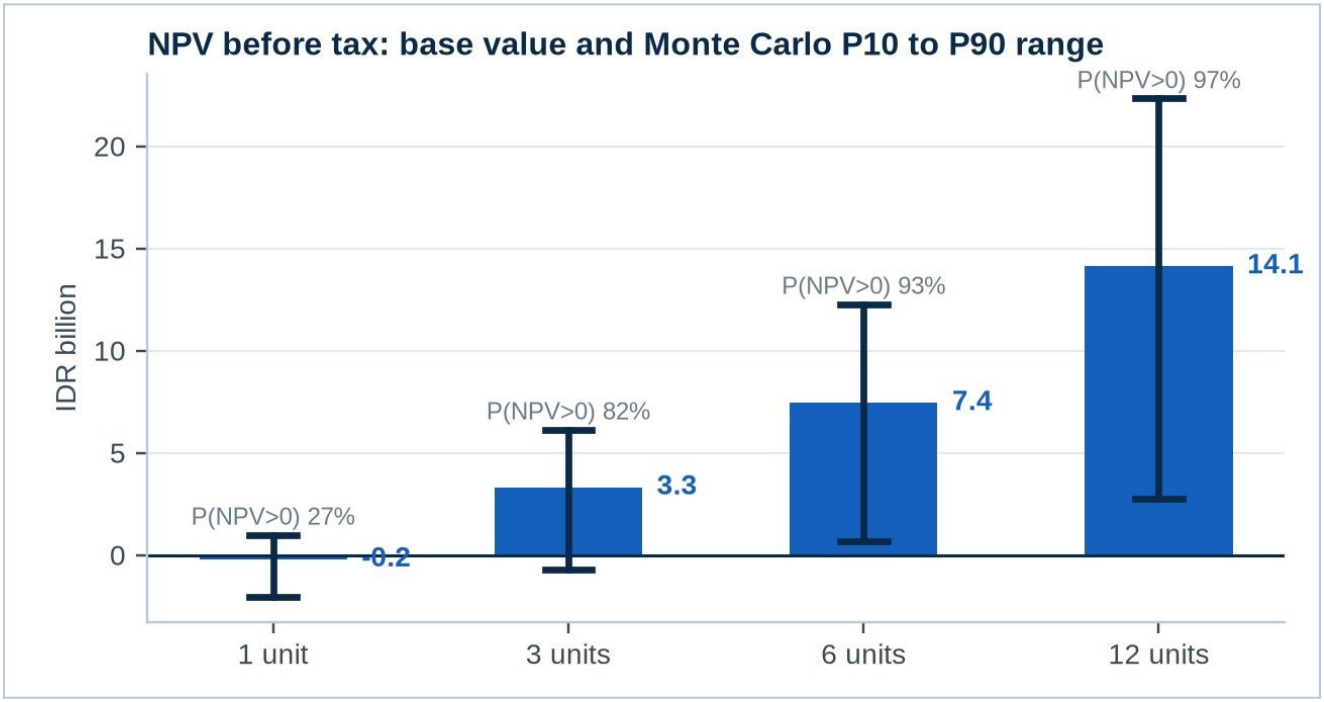


Figure 13. NPV before tax with the Monte Carlo 10th to 90th percentile range. Source: MonteCarlo sheet.

The Monte Carlo run draws 17 drivers independently from ranges that are wider than the scenario switch. For three units the mean NPV is IDR 2,443 M, the median is IDR 2,066 M, and 82% of outcomes are positive. The 5th

percentile is IDR -1,418 M. A single unit is positive in only 27% of outcomes, which confirms that the pilot should not be judged on its own return.

Risk register with financial impact (3 units, Base)

NPV impact is computed by shocking the Base scenario in the Python replica. Grey cells are static.

ID	Risk	Likelihood	NPV after shock (IDR M)	Change vs base (IDR M)	Shock applied in the model
R1	Low adoption by operators and engineers	Medium	255	(3,065)	Steady-state adoption falls by 30 percent
R2	Rollout delayed by one year	Medium	1,753	(1,567)	Units 2 and 3 go live one year later
R3	One-off cost overrun	Medium	2,551	(769)	One-off cost 30 percent higher
R4	Data quality is worse than expected	High	1,825	(1,494)	Ingestion effort doubles and benefits fall by 10 percent
R5	AI service unavailable or withdrawn	Low	2,501	(819)	Search time reduction falls by 30 percent
R6	Benefit attribution is challenged	Medium	1,187	(2,133)	Share credited to the platform falls by 30 percent
R7	Polyethylene margin downturn	Medium	766	(2,554)	Contribution margin falls to USD 140 per tonne
R8	Key person leaves the knowledge manager role	Medium	1,354	(1,966)	Adoption falls by 15 percent and running cost rises by 10 percer
R9	Cybersecurity incident or audit finding	Low	3,020	(300)	Extra one-off cost of IDR 300 M
R10	Data protection or contract constraints	Low	2,917	(403)	Connector cost 50 percent higher and security cost doubled
	Combined stress: R1, R2, R3 and R6 together		(2,643)	(5,963)	

Figure 14. Risk register with the NPV effect of each shock, Risks sheet.

The most damaging single risks are low adoption (NPV IDR 255 M), a challenge to benefit attribution and a margin downturn. A combination of low adoption, a one-year delay, a 30 percent cost overrun and weaker attribution gives an NPV of IDR -2,643 M. The mitigations are role-based onboarding, baselines and control units, and the stage gates in Section 8.

8 KPIs and benefits realisation

KPI	Baseline	Target	Measured by
Mean time to repair	14.0 h	12.6 h	CMMS work order dates
Jobs repeating within 120 days	20.6%	14.4%	Failure memory report
Minutes a day searching and checking	40	20	Pilot survey and timed tasks
Median time to a cited answer	About 15 min	3 min or less	Timed pilot tasks
Answers with an approved source	Not measured	At least 90%	Answer audit log
Weekly active users	0	85% of users	Usage log
Documents without approval status	67%	10% or less	Documents and Data Quality pages

- **Gate 1, month 6.** Measured repair time, search time and adoption replace the assumptions. Continue to unit 2 only if the weighted benefit is at least 70 percent of the Base case.
- **Gate 2, month 12.** Repeat failure share and answer audit reviewed. Approve units 3 and beyond against the sensitivity table.
- **Gate 3, month 24.** Full benefit review, model update and rollout plan.

9 Conclusions and next steps

- The investment case is positive for two or more units and strong at scale: NPV of IDR 3.3 B at three units and IDR 14.1 B at twelve.

- The pilot is a learning investment. Its purpose is to replace assumptions with measured Company data before the programme grows.
- The result depends most on the value of a downtime hour. The first pilot task should be to measure how many breakdown hours reduce output at the real unit.
- The platform is priced for the value it creates inside the Company. External sales are an option once the pilot has proved the numbers.

Decisions requested: approve the pilot budget of IDR 1,766 M with the three gates above; nominate a knowledge manager and a unit sponsor; and give read access to EDMS, AIMS, CMMS and historian extracts for the pilot unit.

9.1 Limits

The work orders and plant data are sample data prepared for the case. Salaries, margins, production loss and cost lines are estimates until the pilot replaces them. The model has no terminal value, financing or working capital, and the Monte Carlo run treats drivers as independent. Market figures come from public summaries of analyst reports and should be read as orders of magnitude.

9.2 Workbook guide and sources

The workbook Business_Impact_Financial_Model.xlsx contains the sheets Inputs, Case_Data, Case_Metrics, Labour, Downtime_Value, Levers, Costs, Rollout, CF_1U to CF_12U, Summary, Market, Competitors, Sensitivity, MonteCarlo, Risks, KPIs, Reconciliation, Checks and Sources. The Sources sheet lists the references with links. Blue cells are inputs, black cells are formulas and grey cells are static outputs of the Python replica.

10 Appendix: assumptions that matter most

> Twelve inputs carry most of the uncertainty. Each is tagged by the quality of its evidence and is replaced by measured data in the pilot.

Assumption	Base	Scenario range	Evidence tag
Minutes a day spent searching and checking	40 min	36 min to 44 min	Assumption
Reduction in search time	50%	45% to 55%	Assumption
Share of freed time that is productive	60%	55% to 65%	Assumption
Steady-state adoption	85%	80% to 90%	Assumption
Reduction in mean time to repair	10.0%	8.5% to 11.5%	Assumption
Share of the gain credited to the platform	70%	65% to 75%	Assumption
Reduction in repeat failures	30%	25% to 35%	Assumption
Contribution margin lost per tonne	USD 220	USD 190 to USD 250	Estimate
Share of breakdown hours with output loss	20%	17% to 23%	Assumption
Unit capacity (kt a year)	336	336 to 336	Market source
New joiners and role changes a year	6	5 to 7	Assumption
Platform configuration (person-months)	8	9 to 7.5	Estimate

Reconciliation to the earlier business impact slide

Earlier values are the figures shown on the previous slide (static, blue). New values are live links. Values at steady-state adoption.

Item	Earlier slide	New model	Change	What changed and why
Users per unit	45	45	0%	Unchanged: 45 users in five role groups.
Search time lever, 1 unit (IDR M a year)	604	340	-44%	Loaded cost now built from role salaries, only 60 percent of freed time counts, and adoption is 85 percent.
Repair time lever, 1 unit (IDR M a year)	857	528	-38%	Downtime value is now built bottom-up. Only 70 percent of the gain is credited and repeat failures are removed first.
Repeat failure lever, 1 unit (IDR M a year)	-	359 new		New lever, based on the 120-day repeat rate in the work order data.
Ramp-up lever, 1 unit (IDR M a year)	60	47	-21%	Now built from joiners, days saved and loaded cost instead of a flat assumption.
Total value, 1 unit (IDR M a year)	1,521	1,275	-16%	Sum of the four headline levers at steady-state adoption.
Value of one breakdown hour (IDR M)	30.0	33.1	10%	Capacity, margin, exchange rate and share of hours with output loss.
Share of freed search time that is productive	100%	60%	-40%	New haircut.
Share of repair gain credited to the platform	100%	70%	-30%	New haircut.
Pilot one-off cost (IDR M)	1,400	1,766	26%	Cost anchor rebuilt line by line on the Costs sheet.
Pilot running cost (IDR M a year)	370	731	98%	Cost anchor rebuilt line by line on the Costs sheet.
Payback, 1 unit (months)	20.9	49.9	138%	Higher cost and ramp-up. A single unit does not repay quickly.
Value a year at 3 units (IDR M)	4,562	3,825	-16%	Year 0 prices at full rollout.
Payback at 3 units (months)	11.8	33.9	186%	Phased rollout and adoption ramp are now modelled by cohort.
Net benefit over 3 years at 3 units (IDR M)	7,930	375	-95%	Earlier slide: net over three years. New: cumulative net cash flow to year 3.
Value a year at 6 units (IDR M)	9,123	7,458	-18%	Year 0 prices at full rollout.
NPV at 3 units (IDR M)	not shown	3,320		New: discounted at the weighted average cost of capital.
IRR at 3 units	not shown	44%		New.
Repeat failures within 120 days	8 of 55	13 of 63		Rule restated: same equipment and same failure mode in the previous 120 days, on all 63 corrective or breakdown jobs.

Figure 15. Reconciliation sheet: how each value on the earlier slide changed and why.